Effective Archimedean Stark Theorems over Real Quadratic Fields:
Quadratic Support, Shintani Transfer, and CM Descent

Hainan Zhao
Independent Researcher
hainzhao@gmail.com

30 July 2026

Abstract

We prove a uniform formula for every one-place real-quadratic Stark invariant whose differenced Fourier support is quadratic. The formula is an explicit product of relative units, with rational exponents computed from class numbers, imprimitive Euler factors, and an exact relative-regulator index. Beyond this quadratic floor, we translate Shintani's index-two theorem into the one-place notation, give a general all-embeddings height-rigidity criterion, and use them to identify eight explicit packets with support orders up to ten. The examples include two order-six packets, an order-six packet at a ramified 3-power conductor, and an order-ten packet. We also prove a cyclic-quartic Fourier-to-CM-norm theorem from Stark's imaginary-quadratic rank-one theorem and apply it to five primitive quartic packets, including examples with $e = |\mu(E)| = 2, 4, 6, 8$. The $e = 8$ route proves the packet over $\mathbb{Q}(\sqrt{6})$; a separate $e = 12$ computation is retained only as a cross-check because its auxiliary-prime valuation argument is not used in the proof. Two structural lemmas prove that an absolutely abelian one-place ray field has trivial differenced support and that nontrivial support forces even Shintani index. The latter is independently consistent with all 446 genuinely reconstructed odd-index cases in a separate census. Exact and Arb certificates accompany the written proofs; no unproved Stark conjecture or floating-point recognition enters a proof.

1 Introduction and claim boundary

Let K be real quadratic and let $\mathfrak{m} = \mathfrak{f}\infty_2$ be a ray modulus containing one real place. For $K = \mathbb{Q}(\sqrt{D})$ with $D > 0$, our convention is that ∞_2 is the negative-square-root embedding $\sqrt{D} \mapsto -\sqrt{D}$, which is the first real place in the pinned PARI ordering; ∞_1 is the positive-square-root embedding. For $A \in \mathrm{Cl}_{\mathfrak{m}}(K)$, let R be the sign class and put

$$Z_{\mathfrak{m}}(s, A) = \zeta_{\mathfrak{m}}(s, A) - \zeta_{\mathfrak{m}}(s, RA), \qquad X_A = \exp Z'_{\mathfrak{m}}(0, A). \tag{1}$$

The numbers X_A are positive real analytic invariants. We call the finite Artin-indexed tuple

$$\mathcal{X}_{\mathfrak{m}} = (X_A)_{A \in \mathrm{Cl}_{\mathfrak{m}}(K)}$$

the *one-place packet*; repeated values may be suppressed when a smaller Artin orbit is displayed. The word "packet" carries no algebraicity assumption. Stark's rank-one conjecture predicts algebraic units with the prescribed Artin action, but that conjecture is not known in this generality. The purpose of this paper is to identify explicit packets only where a proved theorem already supplies algebraicity and where the remaining analytic-to-algebraic comparison can be certified.

The word *unconditional* has this precise meaning throughout: each displayed identity follows from a published proved theorem under hypotheses checked exactly in the case bundle. It does not mean that the general real-quadratic rank-one Stark conjecture is proved. Shintani's theorem is used only in its index-two range [1], and the quadratic cases use the analytic class-number formula. CM descent uses Stark's proved theorem for abelian extensions of imaginary quadratic fields [2]. Numerical PARI recognition, when retained as a cross-check, is outside the proof chain.

Terminology and quantitative conventions

Fourier inversion of (1) involves precisely the ray characters χ for which $\chi(R) = -1$. The *support order* is the maximum order of such a character. Labels such as `RQ-000190` are stable locators in the accompanying corpus; they are not mathematical notation, and every use below also states the field and modulus.

If θ is a supported quartic character and $U = \ker\theta$, its *primitive quartic packet* is the geometric fiber mean

$$Y_{\bar{A}} = \left(\prod_{B\in U} X_{AB}\right)^{1/|U|}, \qquad \bar{A} \in G/U. \tag{2a}$$

It is a positive analytic invariant indexed by $G/U \simeq C_4$. This definition is essential: it removes every Fourier component that does not factor through the selected quartic quotient. When the full differenced support is the inverse pair $\{\theta, \bar{\theta}\}$, X_A is constant on U, and $Y_{\bar{A}} = X_A$.

For Engine B, a *safe exponent* is an explicitly computed integer m for which Shintani's theorem proves that every X_A^m is algebraic in the asserted ray field. If the analytic and algebraic logarithms differ by balls of radius at most δ, the height comparison bounds the quotient by $m\delta$. The reported *margin* is

$$\mathcal{M} = \frac{V_{\min}}{m\delta}, \tag{2}$$

where $V_{\min}$ is the minimum applicable Voutier lower bound over the certified degree window. Thus $\mathcal{M} > 100$ is the pre-registered acceptance condition. A *cone evaluator* is the finite Shintani–Yamamoto decomposition of a partial zeta derivative into double-sine or Mellin-integral terms; all such terms below are evaluated as Arb balls.

How the results fit together

There are three theorem mechanisms. Engine A is a uniform closed-form formula for quadratic Fourier support. Engine B reaches higher character orders when Shintani's maximal absolutely abelian subfield has index two: Proposition 7 translates the algebraicity theorem and Lemma 8 turns certified archimedean comparisons into exact identities. The order-six cases on either side of ramification isolate the index/wildness condition, rather than character order, as the relevant obstruction. Engine C passes from a quartic quotient packet to an abelian cyclic-quartic extension of an imaginary quadratic field. Exact linear reinduction identifies the L-function, and Theorem 9 converts Stark's proved unit into the Artin-labeled positive packet. A packet proved by both Engines B and C tests the overlap of the two theorem bases. Lemmas 10 and 11 explain two further structural boundaries.

Main theorems

Theorem 1 (Uniform quadratic-support theorem)**.** *For every real quadratic K, every one-place modulus $\mathfrak{m} = \mathfrak{f}\infty_2$, and every packet whose differenced Fourier support consists only of quadratic characters, each X_A is an explicit rational power of a product of relative units. Corollary 5 gives one closed product formula whose exponents are expressed entirely in terms of exact class* *numbers, roots of unity, imprimitive Euler factors, and relative-regulator indices. No numerical approximation is required to state or prove the formula.*

Theorem 2 (Selected higher-order packets)**.** *Use the integral basis* $1, \omega$, *where* $\omega = (1+\sqrt{D})/2$ *for* $D \equiv 1 \pmod 4$ *and* $\omega = \sqrt{D}$ *otherwise, and let the displayed HNF matrix denote the finite ideal* $\mathfrak{f}$ *in that basis. For every row below, the packet* $\mathcal{X}_{\mathfrak{f}\infty_2}$ *is the Artin-labeled set of positive roots of the relative polynomial printed in Section 4 or 6.*

case	K	HNF($\mathfrak{f}$)	*support*	m	*margin*
RQ-000190	$\mathbb{Q}(\sqrt{7})$	$\begin{bmatrix} 7 & 0 \\ 0 & 1 \end{bmatrix}$	$2, 6$	*4032*	> 5688
RQ-000419	$\mathbb{Q}(\sqrt{14})$	$\begin{bmatrix} 7 & 0 \\ 0 & 1 \end{bmatrix}$	$2, 6$	*4032*	> 7315
RQ-000108	$\mathbb{Q}(\sqrt{5})$	$\begin{bmatrix} 15 & 6 \\ 0 & 3 \end{bmatrix}$	4	*2880*	> 2460
RQ-000021	$\mathbb{Q}(\sqrt{2})$	$\begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix}$	$2, 6$	*2016*	> 4261
RQ-002057	$\mathbb{Q}(\sqrt{57})$	$\begin{bmatrix} 9 & 3 \\ 0 & 3 \end{bmatrix}$	$2, 6$	*2592*	> 748
RQ-002955	$\mathbb{Q}(\sqrt{77})$	$\begin{bmatrix} 7 & 3 \\ 0 & 1 \end{bmatrix}$	$2, 6$	*4032*	> 5151
RQ-001107	$\mathbb{Q}(\sqrt{33})$	$\begin{bmatrix} 11 & 5 \\ 0 & 1 \end{bmatrix}$	$2, 10$	*15840*	> 5817
RQ-000458	$\mathbb{Q}(\sqrt{14})$	$\begin{bmatrix} 12 & 0 \\ 0 & 6 \end{bmatrix}$	4	*1152*	> 6470

Here m *is the safe exponent of Proposition 7, and "margin" has the meaning in (2). Each row includes exact ray-field identification, unconditional class and unit groups, Sturm isolation, all-embeddings Arb bounds, and exact Artin labels.*

Theorem 3 (Selected cyclic-quartic CM packets)**.** *Use the same integral-basis convention as in Theorem 2. For every row below, the primitive quartic packet (2a) is the Artin-labeled set of positive numbers whose polynomial is printed in Section 6.*

case	K	HNF($\mathfrak{f}$)	*CM bases*	*e used in proof*	*polynomial*
RQ-001280	$\mathbb{Q}(\sqrt{35})$	$\begin{bmatrix} 8 & 4 \\ 0 & 4 \end{bmatrix}$	$\mathbb{Q}(\sqrt{-10}), \mathbb{Q}(\sqrt{-14})$	$2, 2$	P_{35}
RQ-001569	$\mathbb{Q}(\sqrt{42})$	$\begin{bmatrix} 6 & 0 \\ 0 & 2 \end{bmatrix}$	$\mathbb{Q}(\sqrt{-7}), \mathbb{Q}(\sqrt{-6})$	$6, 6$	P_{42}
RQ-007519	$\mathbb{Q}(\sqrt{186})$	$\begin{bmatrix} 6 & 0 \\ 0 & 2 \end{bmatrix}$	$\mathbb{Q}(\sqrt{-6}), \mathbb{Q}(\sqrt{-31})$	$6, 6$	P_{186}
RQ-001894	$\mathbb{Q}(\sqrt{51})$	$\begin{bmatrix} 15 & 0 \\ 0 & 5 \end{bmatrix}$	$\mathbb{Q}(\sqrt{-15}), \mathbb{Q}(\sqrt{-85})$	$6, 6$	P_{51}
RQ-000129	$\mathbb{Q}(\sqrt{6})$	$\begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix}$	$\mathbb{Q}(\sqrt{-2})$	8	P_6

In each route the relative extension over the imaginary quadratic base is cyclic quartic, the exact character identity includes all imprimitive Euler factors, Stark's global-unit hypothesis $|S| \geq 3$ *holds, and the free unit supplied by Stark is certified to be an* $e/2$*-th power modulo torsion. Exact common-normal-closure identities implement Theorem 9. For RQ-000129 only the* $e = 8$ *route is used in this theorem; the* $e = 12$ *auxiliary-prime calculation is not part of the proof.*

Historical scope. We are unaware of earlier unconditional oriented examples with support order six or ten. This historical remark is not part of either theorem and makes no universal priority claim. The public supplement will include the dated search perimeter—sources, versions, hashes, and date—on which the remark is based.[1]

Theorems 2 and 3 are deliberately selected-results theorems, not census-completeness statements. Counts, conductor trends, and a complete frontier taxonomy require the separately audited census and will appear elsewhere. This separation prevents provisional classification data from entering a theorem proof. The prior numerical and conditional computations of Dummit–Sands–Tangedal and Cohen–Roblot remain prior work rather than claims displaced here [6, 7, 8, 9]. More precisely, the 1997 Dummit–Sands–Tangedal computations concern totally real cubic bases and are numerical; their 2003 exponent-two theorems overlap the theorem class of Engine A, which is consequently not advertised as a historical first. Cohen–Roblot construct Hilbert class fields conditionally from Stark units and verify the resulting fields independently, whereas every promoted object here is a one-place ray field with nontrivial finite modulus. Thus these are mathematically adjacent precedents, not identical class-field objects. The order-six and order-ten historical statements refer only to the dated search described above.

[1] Kopp's arXiv:2411.06763 was checked again on 30 July 2026; the current version remains v3, dated 3 May 2025, and its algebraicity applications are conditional.

2 The quadratic-support theorem

Let $G = \mathrm{Cl}_{\mathfrak{m}}(K)$, and suppose every character χ with $\chi(R) = -1$ has order two. Write χ° for the associated primitive Hecke character and L_χ/K for the quadratic extension it cuts out. The selected real place becomes complex in L_χ, while the other real place splits, so L_χ has signature $(2,1)$.

Let ϵ_K generate $E_K/\mu(K)$. Modulo torsion, use the logarithmic embedding

$$\lambda_L(u) = \big(\log|\tau_1 u|, \log|\tau_2 u|, 2\log|\tau_3 u|\big), \tag{3}$$

where τ_1, τ_2 are real and τ_3 is one of the complex embeddings. Let u_χ generate the primitive rank-one kernel of

$$N_{L_\chi/K} : E_{L_\chi}/\mu(L_\chi) \longrightarrow E_K/\mu(K),$$

and orient it by $\log|\tau_1 u_\chi| > 0$. In any exact basis of $E_{L_\chi}/\mu(L_\chi)$, write v_K and v_χ for the coordinate columns of ϵ_K and u_χ, and define

$$I_\chi = [E_{L_\chi}/\mu(L_\chi) : \langle E_K/\mu(K), u_\chi\rangle] = |\det(v_K, v_\chi)|. \tag{4}$$

Proposition 4 (Explicit quadratic formula)**.** *With E_χ the exact product of imprimitive Euler factors at zero,*

$$L'_{\mathfrak{m}}(0,\chi) = E_\chi \frac{h_{L_\chi}}{h_K}\frac{w_K}{w_{L_\chi}}\frac{2}{I_\chi}\log|\tau_1 u_\chi|. \tag{5}$$

Moreover

$$Z'_{\mathfrak{m}}(0,A) = \frac{2}{|G|}\sum_{\chi(R)=-1}\chi(A)^{-1}L'_{\mathfrak{m}}(0,\chi). \tag{6}$$

Consequently a common denominator in (5)–(6) gives an explicit power identity for every X_A.

Proof. For a primitive quadratic χ°, $\zeta_{L_\chi}(s) = \zeta_K(s)L(s,\chi^\circ)$. The analytic class-number formula at zero, with the same regulator normalization as (3), is

$$\zeta_F(s) = -\frac{h_F R_F}{w_F}s^{r_1(F)+r_2(F)-1} + O\Big(s^{r_1(F)+r_2(F)}\Big).$$

Since the unit ranks of K and L_χ are one and two, respectively, division gives

$$L'(0,\chi^\circ) = \frac{h_{L_\chi}}{h_K}\frac{w_K}{w_{L_\chi}}\frac{R_{L_\chi}}{R_K}. \tag{7}$$

We next prove the regulator identity without suppressing the mixed signature. Put $a = \log|\tau_1\epsilon_K|$, where τ_1, τ_2 lie over the split real place of K. The product formula gives

$$\lambda_L(\epsilon_K) = (a, a, -2a).$$

For the relative unit u_χ, exact norm one modulo torsion gives

$$\lambda_L(u_\chi) = (b, -b, 0), \qquad b = \log|\tau_1 u_\chi| > 0.$$

The determinant of the first two coordinates is $2|a|b$. Since $R_K = |a|$, the regulator of $\langle E_K, u_\chi\rangle$ is $2R_K\log|\tau_1 u_\chi|$. The same subgroup has index I_χ in the full free unit group, so its regulator is $I_\chi R_{L_\chi}$. Therefore

$$\frac{R_{L_\chi}}{R_K} = \frac{2}{I_\chi}\log|\tau_1 u_\chi|. \tag{8}$$

If $\mathfrak{m}/\mathfrak{f}_{\chi^\circ}$ is the imprimitive part, then

$$L_{\mathfrak{m}}(s,\chi) = L(s,\chi^\circ) \prod_{\mathfrak{p}\mid\mathfrak{m}/\mathfrak{f}_{\chi^\circ}} \left(1 - \chi^\circ(\mathfrak{p}) N\mathfrak{p}^{-s}\right).$$

At $s = 0$ the product is the exact integer E_χ. If one factor vanishes, both this product and $L(s,\chi^\circ)$ vanish at zero, so $L'_{\mathfrak{m}}(0,\chi) = 0$; otherwise combining (7) and (8) proves (5).

Finally, character inversion gives

$$\zeta_{\mathfrak{m}}(s,A) = \frac{1}{|G|} \sum_{\chi\in\widehat{G}} \chi(A)^{-1} L_{\mathfrak{m}}(s,\chi).$$

Subtracting the formula with A replaced by RA multiplies the χ-term by $1-\chi(R)^{-1}$, which is two precisely for $\chi(R) = -1$ and zero otherwise. Differentiation proves (6). Every surviving character is quadratic, hence every coefficient is rational. If q clears their denominators, exponentiation yields

$$X_A^q = \prod_{\chi(R)=-1} |\tau_1 u_\chi|^{n_{A,\chi}} \qquad (n_{A,\chi} \in \mathbb{Z}), \tag{9}$$

after the exact factors in (5) have been absorbed into the integers $n_{A,\chi}$. This proves Proposition 4 and Theorem 1. □

Corollary 5 (Closed packet formula)**.** *Under the hypotheses of Theorem 1, put*

$$c_{A,\chi} = \frac{4}{|G|} \chi(A)^{-1} E_\chi \frac{h_{L_\chi}}{h_K} \frac{w_K}{w_{L_\chi}} \frac{1}{I_\chi}.$$

Then the positive real invariant at the selected embedding is

$$\boxed{X_A = \prod_{\chi(R)=-1} |\tau_1 u_\chi|^{c_{A,\chi}}.}$$

Every exponent $c_{A,\chi}$ is rational and is computed by exact finite arithmetic.

Remark 6. *The word "uniform" refers to this single character product, not to one field-independent named unit. For fixed $(K,\mathfrak{m},A)$, the supported characters form a finite set, and substituting their exact class numbers, Euler factors, unit kernels, and determinant indices reduces the corollary immediately to a concrete product of explicitly computed relative units.*

The theorem is uniform; its proof does not enumerate the census. There are 1,560 nontrivial eligible occurrences in the frozen range, involving 2,232 supported character occurrences and 912 distinct quartic fields. These figures describe a verification queue, not a completeness claim here. Each application still certifies its primitive conductor, field, class numbers, Euler factors, exact norm map, primitive kernel, determinant, and orientation. The dimension-four and dimension-eight anchors independently replay (5), including relative index $I_\chi = 2$ in both dimension-eight cases.

3 Shintani transfer and height rigidity

Let $H = H_{\mathfrak{m}}$ be the one-place ray class field of $\mathfrak{m} = \mathfrak{f}\infty_2$, and let $\mathcal{N}$ be its normal closure over $\mathbb{Q}$. Engine B first checks Shintani's conditions (0-3), (0-6), and (0-9), including

$$[H : H \cap \mathbb{Q}^{\mathrm{ab}}] = 2. \tag{10}$$

The maximal absolutely abelian subfield is the fixed field of $[\mathrm{Gal}(\mathcal{N}/\mathbb{Q}), \mathrm{Gal}(\mathcal{N}/\mathbb{Q})]$. The imaginary quadratic field and transfer conductor below are then reconstructed a second time from their intrinsic ray class fields.

More precisely, an Engine-B transfer route consists of an imaginary quadratic subfield $k \subset \mathcal{N}$, a finite modulus $\mathfrak{c}$ of k, and a k-isomorphism

$$\iota : H_{\mathfrak{c}}(k) \xrightarrow{\sim} \mathcal{N}.$$

Here $\mathcal{N}/k$ is abelian, $\mathfrak{c} = \mathrm{cond}(\mathcal{N}/k)$, and $H_{\mathfrak{c}}(k)$ denotes the corresponding ray class field. Thus "transfer conductor" below means this Artin conductor, not a conductor guessed from a discriminant. In every selected case it is computed by `rnfconductor` and the isomorphism is independently constructed by `bnrclassfield`; the two routes printed for a case must produce the same $\mathcal{N}$.

Proposition 7 (Shintani transfer in one-place notation)**.** *Let $C = \mathrm{Cl}_{\mathfrak{f}\infty_1\infty_2}(K)$, and suppose exact fiber identities identify Shintani's invariant $X_{\mathfrak{f}}(c, G)$ with X_A in (1). Assume Shintani's conditions (0-3), (0-6), and (0-9), and let k and $\mathfrak{c}$ be respectively the imaginary quadratic field and finite transfer conductor occurring in his proof.*

For $\mathfrak{d} \mid \mathfrak{c}$, let

$$h(\mathfrak{d}) = |\,\mathrm{Cl}_{\mathfrak{d}}(k)|, \qquad w(\mathfrak{d}) = |\{\zeta \in \mu(k) : \zeta \equiv 1 \pmod{\mathfrak{d}}\}|,$$

put

$$n(\mathfrak{d}) = w(\mathfrak{d})\frac{h(\mathfrak{c})}{h(\mathfrak{d})}, \tag{11}$$

and, for $\mathfrak{d} \neq \mathcal{O}_k$, let $f_{\mathfrak{d}}$ be the positive generator of $\mathfrak{d} \cap \mathbb{Z}$. If r clears the explicitly computed real-side distribution and induction denominators, then

$$\begin{aligned} m_{\mathcal{O}_k} &= 12h_k n(\mathcal{O}_k), \\ m_{\mathfrak{d}} &= 12f_{\mathfrak{d}} n(\mathfrak{d}) \quad (\mathfrak{d} \neq \mathcal{O}_k), \\ m &= \mathrm{lcm}\big(r,\, (m_{\mathfrak{d}})_{\mathfrak{d}|\mathfrak{c}}\big) \end{aligned} \tag{12}$$

has the following property:

$$X_A^m \in \mathcal{O}_H^\times, \qquad (X_A^m)^{\mathrm{Art}(B)} = X_{AB}^m. \tag{13}$$

At every embedding of H above the nonsplit real place, $|X_A^m| = 1$.

Proof. Here and in this paragraph, unqualified theorem, proposition, and equation numbers refer to Shintani's paper [1]. His definition (0-5) on p. 140 introduces the positive invariant $X_{\mathfrak{f}}(c)$, and Theorem 1 on p. 140 identifies its logarithm with the difference of the two partial zeta derivatives. Definition (18) on p. 150 forms the distribution-corrected invariant $Y_{\mathfrak{f}}(c)$; definitions (19)–(20) on pp. 151–152 then take the products over G. The assumed exact fiber identity identifies our X_A with $X_{\mathfrak{f}}(c, G)$, while the explicit real-side distribution calculation, whose denominators divide r, passes from $X_{\mathfrak{f}}(c, G)$ to $Y_{\mathfrak{f}}(c, G)$. Under (0-3), (0-6), and (0-9), Shintani's Theorem 2 on p. 141 supplies the index-two algebraicity construction.

On the imaginary-quadratic side, Shintani's equation (8) on p. 146 defines

$$n(S) = w(\mathfrak{f}(S))\frac{|H_k(\mathfrak{c})|}{|H_k(\mathfrak{f}(S))|}.$$

Reindexing the subsets by the divisors $\mathfrak{d} = \mathfrak{f}(S)$ of $\mathfrak{c}$ gives exactly (11). For $\mathfrak{d} \neq \mathcal{O}_k$, equation (4) on p. 143 contains $|\varphi_{\mathfrak{d}}|^{1/(6f_{\mathfrak{d}})}$, and equation (9) on p. 146 contributes the further exponent $1/n(\mathfrak{d})$. Replacing an absolute value by its product with its complex conjugate contributes a factor two. Hence $12 f_{\mathfrak{d}} n(\mathfrak{d})$ clears that row. At the trivial divisor, equation (6) on p. 144 replaces $f_{\mathfrak{d}}$ by h_k, giving $12 h_k n(\mathcal{O}_k)$.

Shintani's Proposition 4 on pp. 154–156 expresses $Y_{\mathfrak{f}}(c, G)$ as an integral product of ratios of the $W_{\mathfrak{c}}$-invariants and introduces no additional rational power. The integer r in (12), which is external to Shintani's imaginary-side construction, clears the explicitly recorded real-side fiber, distribution, and induction denominators. Thus (12) clears every exponent in his product. Finally, Shintani's Proposition 5(i)–(iii) on pp. 156–158 supplies, respectively, the unit and field assertion, Artin covariance, and modulus one under embeddings inducing the nontrivial real automorphism. This proves (13). $\square$

For the headline case $K = \mathbb{Q}(\sqrt{7})$, $\mathfrak{f} = \left[\begin{smallmatrix} 7 & 0 \\ 0 & 1 \end{smallmatrix}\right]$, the short transfer route has $k = \mathbb{Q}(i)$ and $\mathfrak{c} = (7)$. The complete divisor table is

$\mathfrak{d}$	$h(\mathfrak{d})$	$w(\mathfrak{d})$	$n(\mathfrak{d})$	$f_{\mathfrak{d}}$	$m_{\mathfrak{d}}$
$\mathcal{O}_k$	1	4	48	1	576
(7)	12	1	1	7	84

(14)

and the real-side indices are 6 and 1. Hence $\operatorname{lcm}(576, 84, 6) = 4032$. This is a complete instance of Proposition 7; the longer $\mathbb{Q}(\sqrt{-7})$ reconstruction gives the same normal closure and safe power independently.

Lemma 8 (All-embeddings height rigidity)**.** *Let H be a number field of degree D. Suppose $(X_A^m)_A \subset \mathcal{O}_H^\times$ is an algebraic family with Artin covariance, and let $(\alpha_A)_A \subset \mathcal{O}_H^\times$ be an exactly Artin-labeled positive candidate family. For each archimedean place v of H, choose an embedding $\sigma_v : H \hookrightarrow \mathbb{C}$ representing v, and suppose either*

$$\left| \frac{1}{m} \log |\sigma_v(X_A^m)| - \log |\sigma_v(\alpha_A)| \right| \le \epsilon_v \tag{15}$$

or both powered absolute values are proved to be one. Put

$$B_A = \frac{m}{D} \sum_{v \mid \infty} n_v \epsilon_v, \qquad n_v = [H_v : \mathbb{R}], \tag{16}$$

with $\epsilon_v = 0$ in the second case. If

$$B_A < \min \left\{ \frac{1}{2} \log \frac{1 + \sqrt{5}}{2}, \min_{3 \le d \le D} \frac{1}{4d} \left(\frac{\log \log d}{\log d} \right)^3 \right\}, \tag{17}$$

then $X_A = \alpha_A$.

Proof. The quotient $\eta_A = X_A^m / \alpha_A^m$ is a unit of H. Its finite-place contribution to the Weil height is zero. Equations (15)–(16) and the definition of logarithmic height give

$$h(\eta_A) = \frac{1}{D} \sum_{v \mid \infty} n_v \max(0, \log |\eta_A|_v) \le B_A.$$

If $d = [\mathbb{Q}(\eta_A) : \mathbb{Q}] \ge 3$ and η_A is not torsion, Voutier's explicit lower bound contradicts (17) [5]. A non-torsion quadratic unit has height at least $\frac{1}{2} \log((1 + \sqrt{5})/2)$, and a rational unit is ± 1. Thus η_A is a root of unity. At the distinguished real embedding the analytic number X_A and the algebraic candidate α_A are positive. Hence $\eta_A = 1$, and $X_A^m = \alpha_A^m$ there; taking the unique positive m-th root proves the claim. $\square$

For every row of Theorem 2, the embeddings over the split real place are all represented by the printed packet roots and the cone enclosures. At the remaining archimedean places, Proposition 7 gives modulus one for X_A^m, and the reciprocal candidate polynomial is checked exactly, after the substitution $T = X + X^{-1}$, to have its remaining roots on the unit circle. Hence these places contribute $\epsilon_v = 0$ in (16).

The algebraic candidates are certified without a GRH assumption: each class and unit computation uses `bnfinit(P,1)`, followed by the full call `bnfcertify(bnf)` returning 1; the quotient-only flag is not used. Exact `rnfconductor` and `bnrclassfield` round trips identify the relative ray fields, Sturm sequences isolate the positive roots, and split-prime Frobenius actions label them.

Effective but not uniformly cheap. The required raw radius is $O(V_{\min}/m)$. Arb precision therefore grows with $\log m$, but conductor size, packet degree, and cone count also grow, and no uniform complexity claim is made. The order-ten case below has $m = 15840$ and remains comfortable; cases with much larger exponents can exceed the fixed per-case resource cap and belong to the frontier. This is a stated limit of Engine B, not evidence against its completed cases.

4 The order-six and order-ten packets

4.1 $\mathbb{Q}(\sqrt{7})$ at the ramified prime above seven

For $K = \mathbb{Q}(\sqrt{7})$ and $\mathfrak{m} = \mathfrak{p}_7\infty_2$, the one-place and two-place ray groups are C_6 and $C_6 \times C_2$. The support orders are two and six and (10) holds. The commutator-fixed calculation produces $\mathbb{Q}(i)$ and $\mathbb{Q}(\sqrt{-7})$, and their intrinsic conductor reconstructions give the same degree-24 normal closure. The $\mathbb{Q}(i)$ route has two divisor rows and safe exponent 4032.

The packet polynomial over K is

$$\begin{aligned} P_7(X) = X^6 &- (6+2\sqrt{7})X^5 + (22+8\sqrt{7})X^4 \\ &- (34+13\sqrt{7})X^3 + (22+8\sqrt{7})X^2 \\ &- (6+2\sqrt{7})X + 1. \end{aligned} \tag{19}$$

It is irreducible over K, defines the one-place ray field, has six positive Sturm-isolated roots, and is Artin-labeled by the split primes 19 and 31.

For the class represented by $(3)^r$, an exact Yamamoto domain has generators

$$\lambda_r = (7-2\sqrt{7})/3^r, \qquad \lambda_r(8+3\sqrt{7}).$$

Its determinant is nine and its half-open parallelotope contains exactly nine lattice points. The resulting 27 independent double-sine evaluations are enclosed without a PARI L-value. At exponent 4032, the powered height upper bound is 9.1903×10^{-9}. The minimum Voutier lower bound in degrees three through 24 is 5.2279×10^{-5}, so the margin exceeds 5688. The degree-one and degree-two fallbacks finish the proof of (19).

This is the headline order-six instance. The same character order is the difficult component at the separate $\mathbb{Q}(\sqrt{21})$ wall, but here the index-two hypothesis holds and the packet closes. Thus the character order itself is not the obstruction.

4.2 Replications and the ramified-prime-3 control

For $\mathbb{Q}(\sqrt{14})$ at $\mathfrak{p}_7$, both $\mathbb{Q}(\sqrt{-7})$ and $\mathbb{Q}(\sqrt{-2})$ reconstruct the degree-24 normal closure. The latter route has clearing exponents 576 and 84, with least common multiple 4032. The packet is

$$P_{14}(X) = X^6 - (13+4\sqrt{14})X^5 + (85+22\sqrt{14})X^4 \\ - (139+38\sqrt{14})X^3 + (85+22\sqrt{14})X^2 \\ - (13+4\sqrt{14})X + 1. \tag{20}$$

Split primes 11 and 103 label the Artin generators. The 20-point cone evaluator gives powered height at most 7.1466×10^{-9}, hence margin greater than 7315.

The strongest boundary control is RQ-002057 over $\mathbb{Q}(\sqrt{57})$, with modulus of norm 27. The conductor is a power of the ramified prime 3 and the support still has order six. The bases $\mathbb{Q}(\sqrt{-3})$ and $\mathbb{Q}(\sqrt{-19})$ independently give the same degree-24 closure. On the shorter route the divisor exponents are $864, 324, 108$, hence $m = 2592$. With $y = (1+\sqrt{57})/2$, so $y^2 - y - 14 = 0$, its packet is

$$P_{57}(X) = X^6 - (30+9y)X^5 + (582+177y)X^4 \\ - (2529+772y)X^3 + (582+177y)X^2 \\ - (30+9y)X + 1. \tag{20a}$$

The fundamental positive unit has order three modulo the finite modulus, so a ray-class domain is the union of three adjacent cones, with 240 exact affine points per class. The powered height is at most 6.982×10^{-8}, giving margin greater than 748. Thus order-six support and ramified-prime-3 structure are both reachable; neither alone explains the separate $\mathbb{Q}(\sqrt{21})$ boundary.

4.3 Three further Shintani closures

The three remaining rows of Theorem 2 use exactly the Engine-B proof above; we print their routes and algebraic candidates so that no theorem row is supported only by a corpus identifier.

For RQ-000108, $K = \mathbb{Q}(\sqrt{5})$ and $\mathrm{N}\mathfrak{f} = 45$. The two imaginary bases are $\mathbb{Q}(\sqrt{-3})$ and $\mathbb{Q}(\sqrt{-15})$, and both reconstruct the same degree-16 normal closure. With $y = (1+\sqrt{5})/2$, the relative packet polynomial is

$$X^4 - (1+6y)X^3 + (9+9y)X^2 - (1+6y)X + 1. \tag{21}$$

It is the certified one-place ray field over K. At $m = 2880$, the powered height is at most 2.12465×10^{-8}, so the Voutier margin exceeds 2460.

For RQ-000021, $K = \mathbb{Q}(\sqrt{2})$ and $\mathfrak{f} = (7)$. The bases $\mathbb{Q}(\sqrt{-7})$ and $\mathbb{Q}(\sqrt{-14})$ reconstruct the same degree-24 closure. The packet polynomial is

$$X^6 - (10+6\sqrt{2})X^5 + (58+40\sqrt{2})X^4 \\ - (129+90\sqrt{2})X^3 + (58+40\sqrt{2})X^2 \\ - (10+6\sqrt{2})X + 1. \tag{22}$$

The cone boundary meets a split-prime face; the canonical value is the average of the upper and lower half-open conventions. Both are enclosed independently. At $m = 2016$, the powered height is at most 1.22671×10^{-8}, giving margin greater than 4261.

For RQ-002955, $K = \mathbb{Q}(\sqrt{77})$ and $\mathfrak{f} = \mathfrak{p}_7$ has HNF $\left[\begin{smallmatrix} 7 & 3 \\ 0 & 1 \end{smallmatrix}\right]$. The bases $\mathbb{Q}(\sqrt{-7})$ and $\mathbb{Q}(\sqrt{-11})$ give the same degree-24 closure. Put $y = (1+\sqrt{77})/2$, so $y^2 - y - 19 = 0$. The packet is

$$X^6 - (15+3y)X^5 + (107+26y)X^4 \\ - (217+54y)X^3 + (107+26y)X^2 - (15+3y)X + 1. \tag{23}$$

At $m = 4032$, the powered height is at most 1.01485×10^{-8}, and the margin exceeds 5151. In all three cases the exact ray-field identification, Sturm isolation, Artin labels, and the degree-one and degree-two fallbacks are included in the cited case records.

4.4 Order ten

For RQ-001107 over $\mathbb{Q}(\sqrt{33})$ and $\mathfrak{p}_{11}$, put $y = (1+\sqrt{33})/2 > 0$, so $y^2 - y - 8 = 0$. The packet is

$$\begin{aligned} P_{33}(X) = X^{10} &- (8+4y)X^9 + (74+30y)X^8 - (294+125y)X^7 \\ &+ (669+281y)X^6 - (871+368y)X^5 + (669+281y)X^4 \\ &- (294+125y)X^3 + (74+30y)X^2 - (8+4y)X + 1. \end{aligned} \tag{24}$$

Its relative degree is ten, so the certified ray field H/K has absolute degree 20. The normal closure used to compute the Frobenius action has degree 40, but the comparison quotient $\eta_A = X_A^m/\alpha_A^m$ lies in H, since both powered terms do. Therefore Lemma 8 needs only $d \le 20$. The archived computation conservatively evaluated every degree through 80; on either window the minimum Voutier bound is

$$5.22795332222684\ldots \times 10^{-5}.$$

At $m = 15840$, the factor-100 raw-error target is $3.30047558221391\ldots \times 10^{-11}$; the achieved bound is 5.6731×10^{-13}, so the certified margin exceeds 5817. The degree-one and degree-two fallbacks are recorded separately. This support order is the first one in the present proved inventory divisible by the prime 5; its historical priority status is the survey-bounded status stated after Theorem 2.

5 Cyclic-quartic CM descent

Let $G = \mathrm{Cl}_{\mathfrak{m}}(K)$, let θ be a supported quartic character, and put $U = \ker\theta$. Fix $\bar{s} \in G/U$ with $\theta(\bar{s}) = i$. By character inversion and the definition (2a),

$$\log Y_{\bar{s}^r} = \frac{1}{2}\left(i^{-r}L'_{\mathfrak{m}}(0,\theta) + i^r L'_{\mathfrak{m}}(0,\bar{\theta})\right) = \Re\left(i^{-r}L'_{\mathfrak{m}}(0,\theta)\right). \tag{25}$$

Indeed averaging over U kills every character nontrivial on U; the only odd characters of $G/U \simeq C_4$ are $\theta, \bar{\theta}$. This is the missing link between an individual one-place packet component and a quartic L'-value.

For Engine C, let k be imaginary quadratic and let E/k be a cyclic quartic extension with

$$C = \mathrm{Gal}(E/k) = \langle\sigma\rangle, \qquad \psi(\sigma) = i.$$

Write $\tau = \sigma^2$ for the internal involution over k. Global complex conjugation is a different automorphism, denoted by j, of the common normal closure, because it acts nontrivially on the imaginary quadratic base.

Theorem 9 (Cyclic-quartic CM norm bridge)**.** *Assume the following exact hypotheses.*

1. *Linear reinduction, including the imprimitive local factors, gives*

$$\mathrm{Ind}_{G_K}^{G_{\mathbb{Q}}}\theta \simeq \mathrm{Ind}_{G_k}^{G_{\mathbb{Q}}}\psi, \qquad L_{\mathfrak{m}}(s,\theta) = L_S(s,\psi). \tag{26}$$

2. *Stark's proved imaginary-quadratic rank-one theorem applies to $(E/k, S)$, with $|S| \ge 3$, and supplies a global unit ε, unique modulo $\mu(E)$. Put $e = |\mu(E)|$.*

3. *In $\mathcal{O}_E^\times/\mu(E)$, $\tau(\varepsilon) = \varepsilon^{-1}$ and $\varepsilon = u^{e/2}$ for a certified free unit u.*

4. *The exact normal-closure embedding identifies $\bar{s}$ with σ, and global complex conjugation satisfies*

$$j(E) = E, \qquad j|_k \ne 1.$$

Thus $j \in \mathrm{Aut}(E)$, and the embedding identifies this automorphism with complex conjugation on E.

Put $E^+ = E^{\langle j \rangle}$. Then E/E^+ is quadratic and the norm below is defined. Then, for $r \in \mathbb{Z}/4\mathbb{Z}$,

$$\boxed{Y_{\bar{\mathfrak{s}}^r} = N_{E/E^+}(\sigma^{-r}u)^{-1}.} \tag{27}$$

The right side is independent of the root-of-unity representative of ε. Thus exact character selection, unit divisibility, and the common-normal-closure identities identify the entire Artin-labeled primitive quartic packet.

Proof. Stark's normalized absolute value at the complex place is the square of the ordinary complex modulus. If $\ell_g = \log |g\varepsilon|_{\mathrm{ord}}$, his proved formula therefore reads

$$\zeta_S'(0, g) = -\frac{2}{e}\ell_g. \tag{28}$$

The anti-unit relation gives $\ell_{\sigma^2} = -\ell_1$ and $\ell_{\sigma^3} = -\ell_\sigma$. Fourier transformation with $\psi(\sigma) = i$ gives

$$L_S'(0, \psi) = -\frac{4}{e}(\ell_1 - i\ell_\sigma). \tag{29}$$

These are respectively the class-log coefficient $2/e$ and the direct-L' coefficient $4/e$; their inverse recovery coefficients are $-e/2$ and $-e/4$.

Write $m_r = \log |\sigma^r u|_{\mathrm{ord}}$. The exact divisibility in hypothesis 3 gives $\ell_{\sigma^r} = (e/2)m_r$, so (29) becomes

$$L_S'(0, \psi) = -2(m_0 - im_1).$$

By (26) this is $L_{\mathfrak{m}}'(0, \theta)$. Substitution in (25) yields, successively,

$$(\log Y_1, \log Y_{\bar{\mathfrak{s}}}, \log Y_{\bar{\mathfrak{s}}^2}, \log Y_{\bar{\mathfrak{s}}^3}) = (-2m_0,\ 2m_1,\ 2m_0,\ -2m_1).$$

Since the internal anti-unit relation gives $(m_2, m_3) = (-m_0, -m_1)$, while

$$\log N_{E/E^+}(\sigma^r u) = 2m_r$$

at the selected complex embedding, this vector is exactly the logarithm of the right side of (27). Both sides are positive, so equality of logarithms proves (27). If u is multiplied by a root of unity ξ, then $j(\xi) = \xi^{-1}$, and its CM norm is unchanged. $\square$

Finite character selection and algebraic closure. The hypotheses of Theorem 9 are finite checks. The pipeline enumerates every linear ray character compatible with the relative conductor and the common normal closure. Scalar twists are not permitted: (26) is checked as an equality of ordinary characters on every conjugacy class. Exact Dirichlet coefficients are appended until they separate the finite candidate set. The selected L'-value is enclosed by an independent theta/Mellin evaluation, and Arb inversion in $\mathcal{O}_E^\times/\mu(E)$ must isolate one integral anti-unit orbit. Exact norm identities in the common normal closure then verify (27) at every Artin label. Equality of unlabeled polynomials is never the bridge.

6 CM packet identifications

6.1 The mixed-class-number packet over $\mathbb{Q}(\sqrt{35})$

For RQ-001280 the primitive finite ideal has HNF $\left[\begin{smallmatrix} 8 & 4 \\ 0 & 4 \end{smallmatrix}\right]$ and norm 32. The two exact routes use $\mathbb{Q}(\sqrt{-10})$ and $\mathbb{Q}(\sqrt{-14})$. Their character fields have $e = 2$, their base class numbers are respectively 2 and 4, and $|S| = 3$ in both routes. The exact coefficient selector, Arb unit-orbit isolation, and common-normal-closure identities give

$$\begin{aligned} P_{35}(X) = X^8 &- 38904X^7 + 905404X^6 + 62136X^5 \\ &- 873210X^4 + 62136X^3 + 905404X^2 - 38904X + 1. \end{aligned} \tag{30}$$

It has four positive roots and two nonreal conjugate pairs. Thus the four real roots are precisely the packet (27), while the full selector field has signature $[4, 2]$. Exact conductor transport also proves the member at HNF $\left[\begin{smallmatrix} 8 & 0 \\ 0 & 8 \end{smallmatrix}\right]$ of norm 64.

6.2 The $e = 6$ tranche

The following three cases each have two independent cyclic-quartic routes, $|S| \geq 3$, and the exact divisibility $\varepsilon = u^3$ modulo $\mu(E)$. Theorem 9 therefore applies without a new normalization.

For RQ-001569 over $\mathbb{Q}(\sqrt{42})$, at HNF $\left[\begin{smallmatrix} 6 & 0 \\ 0 & 2 \end{smallmatrix}\right]$, the bases are $\mathbb{Q}(\sqrt{-7})$ and $\mathbb{Q}(\sqrt{-6})$, and

$$\begin{aligned} P_{42}(X) = X^8 &- 416X^7 + 2332X^6 - 2144X^5 \\ &+ 2758X^4 - 2144X^3 + 2332X^2 - 416X + 1. \end{aligned} \tag{31}$$

For RQ-007519 over $\mathbb{Q}(\sqrt{186})$, at the same HNF, the bases are $\mathbb{Q}(\sqrt{-6})$ and $\mathbb{Q}(\sqrt{-31})$, and

$$\begin{aligned} P_{186}(X) = X^8 &- 90243296X^7 + 206015683996X^6 \\ &+ 697017342688X^5 + 984190237126X^4 \\ &+ 697017342688X^3 + 206015683996X^2 - 90243296X + 1. \end{aligned} \tag{32}$$

For RQ-001894 over $\mathbb{Q}(\sqrt{51})$, at HNF $\left[\begin{smallmatrix} 15 & 0 \\ 0 & 5 \end{smallmatrix}\right]$, the bases are $\mathbb{Q}(\sqrt{-15})$ and $\mathbb{Q}(\sqrt{-85})$, and

$$\begin{aligned} P_{51}(X) = X^8 &- 1430858X^7 + 316969003X^6 - 224768756X^5 \\ &+ 477153445X^4 - 224768756X^3 + 316969003X^2 \\ &- 1430858X + 1. \end{aligned} \tag{33}$$

Each polynomial has four positive roots and two nonreal conjugate pairs, and each pair of routes gives identical Artin labels, not just the same polynomial. These three polynomials supersede the larger powered polynomials in the first $e = 6$ bridge record: that record took CM norms of ε, whereas (27) requires the exact cube root u. The correction divides the certified free-unit coordinates by three and reruns both normal-closure routes.

6.3 $\mathbb{Q}(\sqrt{6})$: one proof and one quarantined cross-check

For RQ-000129 the finite ideal has HNF $\left[\begin{smallmatrix} 4 & 0 \\ 0 & 2 \end{smallmatrix}\right]$. The route over $\mathbb{Q}(\sqrt{-2})$ has $e = 8$ and its natural Stark set already has $|S| = 3$. The exact unit lattice gives $\varepsilon = u^4$ modulo $\mu(E)$. Thus Theorem 9 directly proves

$$P_6(X) = X^8 - 8X^7 + 12X^6 + 8X^5 - 10X^4 + 8X^3 + 12X^2 - 8X + 1. \tag{34}$$

Sturm isolation gives four positive roots and no negative roots.

The second route over $\mathbb{Q}(\sqrt{-3})$ has $e = 12$, but its natural set has $|S| = 2$. Enlargements at the rational primes 3 and 5 reproduce the same analytic orbit and the exact identity $q_8^3 = q_{12}^2$. These facts are a strong cross-check, but they are not used in the theorem: the current record does not print the individual auxiliary prime ideals and finite valuations needed to exclude a residual S-unit ambiguity. The earlier zero-real-root polynomial belonged to the complex anti-unit before unit-lattice division and CM norm; it is not a packet polynomial.

6.4 An Engine-B packet and a CM normalization diagnostic

For RQ-000458, $K = \mathbb{Q}(\sqrt{14})$, the finite ideal has HNF $\left[\begin{smallmatrix} 12 & 0 \\ 0 & 6 \end{smallmatrix}\right]$, the ray group is $C_4 \times C_2$, and the support is the inverse quartic pair. The Engine-B proof identifies

$$P_{458}(X) = X^4 - (20 + 6\sqrt{14})X^3 + (138 + 36\sqrt{14})X^2 - (20 + 6\sqrt{14})X + 1. \tag{35}$$

The safe exponent is 1152, and its powered height bound 8.0791×10^{-9} gives margin > 6470. The Engine-C character calculation over $\mathbb{Q}(\sqrt{-42})$ and $\mathbb{Q}(\sqrt{-3})$ independently selects the same quartic L-function, but the banked orientation record uses a fourth-root normalization rather than the $e/2$-root normalization in Theorem 9. Its isolated orbit is

$$(-8, 4), \quad (0, -4), \quad (8, -4), \quad (0, 4),$$

and 32 exact common-closure identities recover (35), but that calculation is retained here only as a diagnostic until its Fourier-scaling convention is rederived. The theorem claim for this row rests solely on Engine B.

7 Structural boundary lemmas

Lemma 10 (Absolutely abelian ray fields)**.** *Let H/K be the one-place ray class field of a real quadratic field. If $H/\mathbb{Q}$ is Galois and abelian, then the sign class $R \in \mathrm{Gal}(H/K)$ is trivial and the differenced Fourier support in (1) is empty.*

Proof. If the one-place field $H/\mathbb{Q}$ were abelian, the complex conjugations above the two real places of K would be conjugate: any lift of the nontrivial element of $\mathrm{Gal}(K/\mathbb{Q})$ interchanges the places. In an abelian group conjugate elements are equal. The inertia element at the omitted real place is trivial in the one-place ray field, whereas the inertia element at the retained place is R. Hence $R = 1$. It follows immediately that no character satisfies $\chi(R) = -1$, so the differenced Fourier support is empty. $\square$

Engine A is not an exception: Proposition 4 reduces individual quadratic characters through (5)–(6), but it does not assert that the entire mixed-signature one-place ray field is absolutely abelian.

Lemma 11 (Index parity)**.** *Let H/K be a one-place ray field and let $N/\mathbb{Q}$ be its genuine normal closure. If the sign class R is nontrivial, then*

$$2 \mid [H : H \cap \mathbb{Q}^{\mathrm{ab}}]. \tag{32}$$

Equivalently, an odd Shintani index can occur only when the Fourier support of (1) is empty.

Proof. Put $G = \mathrm{Gal}(N/\mathbb{Q})$, $A = \mathrm{Gal}(N/K)$, and $B = \mathrm{Gal}(N/H)$. Let $c_1, c_2 \in A$ be the real-place inertia involutions. A lift of the nontrivial automorphism of $K/\mathbb{Q}$ interchanges them, so they are conjugate in G. Hence $c_1 c_2^{-1} \in [G, G]$. Since $G/A \simeq C_2$, one has $[G, G] \subseteq A$. Moreover B is normal in A, so $[G, G]B$ is a subgroup of A. In $A/B = \mathrm{Gal}(H/K)$, the involution at the omitted place is trivial and the one at the retained place is R. Thus the image of $[G, G]$ in A/B contains R. Its order is the degree of H over its fixed field. That fixed field is

$$H \cap N^{[G,G]} = H \cap \mathbb{Q}^{\mathrm{ab}},$$

because $N^{[G,G]}$ is the maximal abelian subextension of $N/\mathbb{Q}$. Hence the image has order $[H : H \cap \mathbb{Q}^{\mathrm{ab}}]$, which is even if $R \neq 1$. Finally, characters of a finite abelian group separate a nonidentity element of order two, so $R = 1$ is equivalent to empty differenced support. $\square$

The use of the genuine normal closure in Lemma 11 is essential. Conjugation coordinates at an unstable modulus do not define $[G, G]$. As an independent consistency check, a pre-registered replay over all 8,200 genuine normal-closure reconstructions found 446 odd indices greater than one. Every one had trivial sign class and empty differenced Fourier support; there were no exceptions.

A Certificates and provenance

The exact field computations use PARI/GP 2.15.4; analytic certificates use Python 3.12.3 and python-flint 0.9.0. The principal replays were run under Linux/x86-64 on an AMD EPYC 9354P virtual CPU. All analytic conclusions are Arb balls, never point estimates. AI-assisted systems proposed code, text, and adversarial checks but were not mathematical authorities: every accepted computation was rerun by the pinned exact or Arb pipeline and reviewed by the author. The disclosure immediately before the bibliography gives the tools and purposes.

The principal machine-readable records are

result	record
first order six	`data/q7-p7-case-v1.json`
second order six	`data/q14-p7-case-v1.json`
RQ-000108	`data/rq000108-case-v1.json`
RQ-000021	`data/rq000021-case-v1.json`
ramified-prime-3 control	`data/q57-norm27-case-v1.json`
RQ-002955	`data/rq002955-case-v1.json`
order ten	`data/q33-p11-order10-case-v1.json`
cyclic-quartic character selection	`artifacts/engine-c-character-selection-v1.json`
$\mathbb{Q}(\sqrt{35})$ CM packet	`artifacts/engine-c-w3-tranche-01-verified-v1.json`
$e=6$ powered-unit record	`artifacts/engine-c-e6-tranche-01-verified-v1.json`
$e=6$ primitive correction	`artifacts/engine-c-e6-primitive-packet-correction-v1.json`
$\mathbb{Q}(\sqrt{6})$, $e=8$ proof route	`data/q6-norm8-case-v3.json`
Engine-C scope correction	`artifacts/engine-c-claim-scope-correction-v1.json`
dual-route packet	`data/rq000458-dual-case-v1.json`
all archimedean places	`artifacts/engine-b-archimedean-place-audit-v1.json`
quadratic theorem	`data/engine-a-uniform-theorem-v1.json`
Shintani source map	`artifacts/shintani-1978-source-map-v1.json`
index-parity theorem	`artifacts/results-paper-index-parity-lemma-v1.json`
parity audit	`artifacts/results-paper-odd-index-parity-audit-v1.json`
literature perimeter	`data/literature-perimeter-v1.json`
Kopp version watch	`artifacts/kopp-arxiv-watch-2026-07-30.json`

Artin-label replay

For each cyclic ray group below, r denotes the power of the generator in the pinned PARI ray coordinates. The entry following r is a rational Sturm interval containing exactly one positive root of the absolute packet polynomial. The cone evaluation for class r, Shintani's Artin covariance (13), and the exact ray-field isomorphism identify that root with X_r.

case	r : isolating interval
RQ-000190	$0:(5.325,5.326)$, $1:(2.713,2.714)$, $2:(2.251,2.252)$, $3:(.187,.188)$, $4:(.368,.369)$, $5:(.444,.445)$
RQ-000419	$0:(20.431,20.432)$, $1:(.189,.190)$, $2:(1.090,1.091)$, $3:(.048,.049)$, $4:(5.289,5.290)$, $5:(.916,.917)$
RQ-000108	$0:(7.892,7.893)$, $1:(2.242,2.243)$, $2:(.126,.127)$, $3:(.445,.446)$
RQ-000021	$0:(6.109,6.110)$, $1:(7.496,7.497)$, $2:(4.351,4.352)$, $3:(.163,.164)$, $4:(.133,.134)$, $5:(.229,.230)$
RQ-002057	$0:(5.709,5.710)$, $1:(.028,.029)$, $2:(.035,.036)$, $3:(.175,.176)$, $4:(34.732,34.733)$, $5:(27.792,27.793)$
RQ-002955	$0:(18.232,18.233)$, $1:(2.180,2.181)$, $2:(8.620,8.621)$, $3:(.054,.055)$, $4:(.458,.459)$, $5:(.116,.117)$
RQ-001107	$0:(8.645,8.646)$, $1:(.276,.277)$, $2:(1.245,1.246)$, $3:(.571,.572)$, $4:(.236,.237)$, $5:(.115,.116)$, $6:(3.611,3.612)$, $7:(.802,.803)$, $8:(1.748,1.749)$, $9:(4.234,4.235)$

Each record carries hashes of its exact and analytic dependencies. The supplementary manifest also records the full software versions, commands, source hashes, and failed computations;

those provenance details are not part of the mathematical argument.

Data availability. This major-revision version is not a finished referee draft until its companion archive has a public immutable identifier. Before such a circulation or any submission, the archive will be frozen and deposited. It will contain the manuscript source, a complete SHA-256 manifest, expected verifier outputs, exact and Arb certificates, and a short README separating discovery programs from proof-checking programs. Separate minimal verification commands will cover the quadratic theorem, every selected Engine-B packet, the cyclic-quartic bridge and selected Engine-C packets, and the two structural lemmas. The environment record will state the software versions, PARI proof modes (including `bnfinit(P,1)` followed by `bnfcertify(bnf)=1`), and expected runtime and memory.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During preparation of this work, the author used OpenAI GPT-5.6 for mathematical brainstorming, symbolic-computation scaffolding, code review, and manuscript editing, and Anthropic Claude for independent adversarial checking of statements and certificates. The author reviewed and verified all outputs, made all mathematical and editorial decisions, and takes full responsibility for the content of the article.

References

[1] T. Shintani, *On certain ray class invariants of real quadratic fields*, J. Math. Soc. Japan **30** (1978), 139–167, `https://doi.org/10.2969/jmsj/03010139`.

[2] H. M. Stark, *L-functions at $s = 1$. IV. First derivatives at $s = 0$*, Adv. Math. **35** (1980), 197–235, `https://doi.org/10.1016/0001-8708(80)90049-3`.

[3] S. Yamamoto, *On Shintani's ray class invariant for totally real number fields*, Math. Ann. **346** (2010), 559–593, `https://doi.org/10.1007/s00208-009-0405-x`.

[4] B. A. Tangedal, *Continued fractions, special values of the double sine function, and Stark units over real quadratic fields*, J. Number Theory **124** (2007), 291–313, `https://doi.org/10.1016/j.jnt.2006.09.011`.

[5] P. Voutier, *An effective lower bound for the height of algebraic numbers*, Acta Arith. **74** (1996), 81–95, `https://doi.org/10.4064/aa-74-1-81-95`.

[6] D. S. Dummit, J. W. Sands, and B. A. Tangedal, *Computing Stark units for totally real cubic fields*, Math. Comp. **66** (1997), 1239–1267, `https://doi.org/10.1090/S0025-5718-97-00852-1`.

[7] D. S. Dummit, J. W. Sands, and B. A. Tangedal, *Stark's conjecture in multiquadratic extensions, revisited*, J. Théorie des Nombres de Bordeaux **15** (2003), 83–97, `https://doi.org/10.5802/jtnb.388`.

[8] H. Cohen and X.-F. Roblot, *Computing the Hilbert class field of real quadratic fields*, Math. Comp. **69** (2000), 1229–1244, `https://doi.org/10.1090/S0025-5718-99-01111-4`.

[9] X.-F. Roblot, *Index formulae for Stark units and their solutions*, Pacific J. Math. **266** (2013), 391–422, `https://doi.org/10.2140/pjm.2013.266.391`.

[10] G. Kopp, *The Shintani–Faddeev modular cocycle: Stark units from q-Pochhammer ratios*, arXiv:2411.06763.

[11] S. Dasgupta and M. Kakde, *On the Brumer–Stark conjecture*, Ann. of Math. **197** (2023), 289–388, `https://doi.org/10.4007/annals.2023.197.1.5`.

[12] B. A. Tangedal and P. T. Young, *Explicit computation of Gross–Stark units over real quadratic fields*, J. Number Theory **133** (2013), 1022–1045, `https://doi.org/10.1016/j.jnt.2012.04.021`.

[13] The PARI Group, *PARI/GP version 2.15.4*, `https://pari.math.u-bordeaux.fr/`.

[14] F. Johansson and contributors, *python-flint 0.9.0*, `https://github.com/flintlib/python-flint`.